

Application of derivative; L'Hôpital's Rule

Consider: $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

If we try to evaluate this by direct substitution, we get: $\frac{0}{0}$

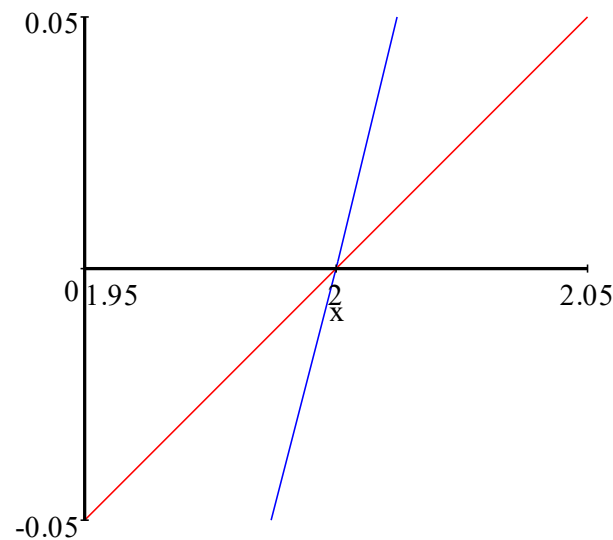
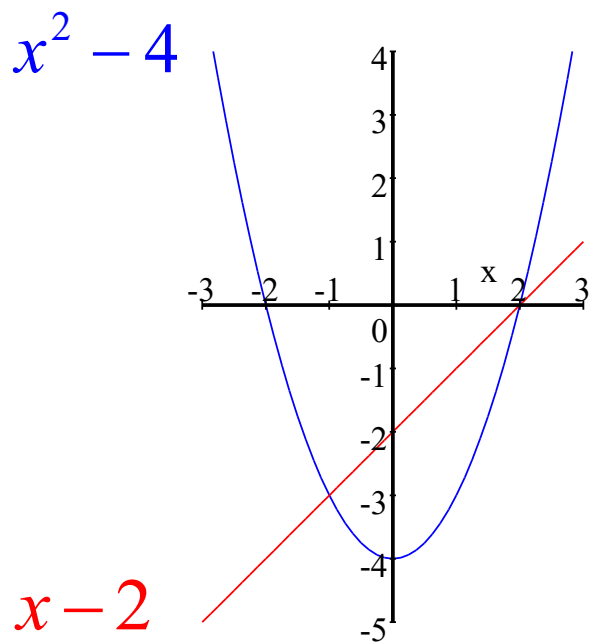
Zero divided by zero cannot be evaluated, and is an example of **indeterminate form**.

In this case, we can evaluate this limit by factoring and canceling:

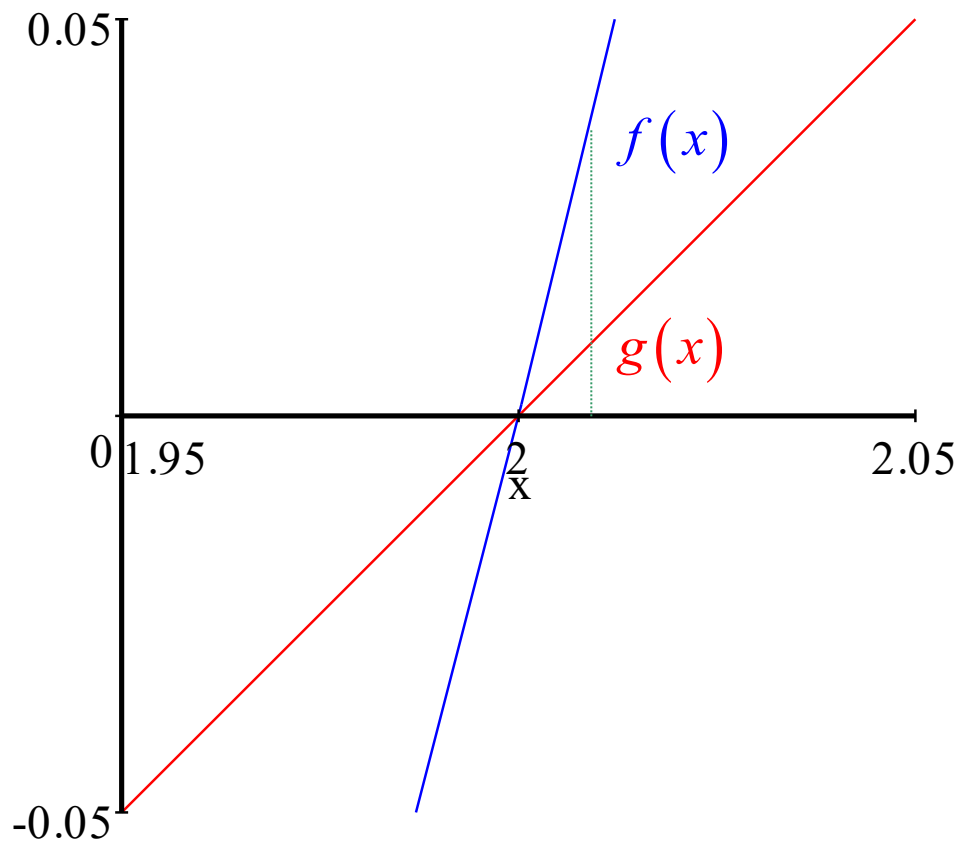
$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x + 2)(\cancel{x - 2})}{\cancel{x - 2}} = \lim_{x \rightarrow 2} (x + 2) = 4$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

The limit is the ratio of the **numerator** over the **denominator** as x approaches 2.



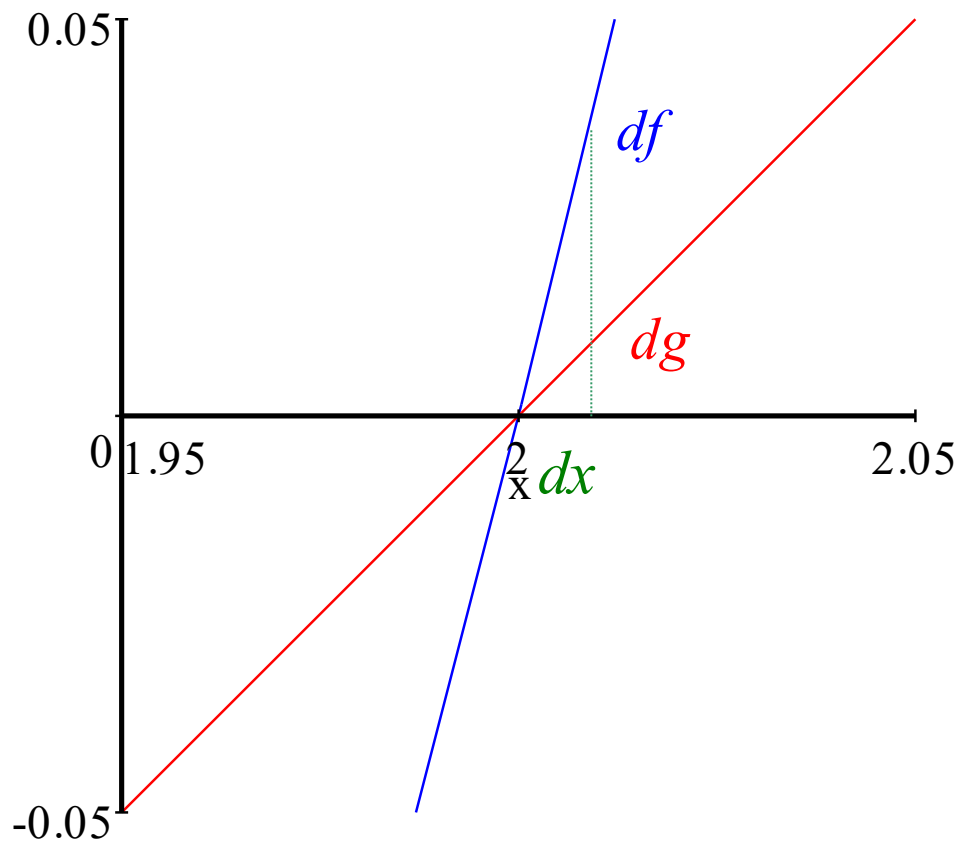
$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$



As $x \rightarrow 2$

$\frac{f(x)}{g(x)}$ becomes:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$



As $x \rightarrow 2$

$\frac{f(x)}{g(x)}$ becomes:

$$\frac{df}{dg} = \frac{\frac{df}{dx}}{\frac{dg}{dx}}$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{\frac{d}{dx}(x^2 - 4)}{\frac{d}{dx}(x - 2)} = \lim_{x \rightarrow 2} \frac{2x}{1} = 4$$

L'Hôpital's Rule:

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is indeterminate, then:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

We can confirm L'Hôpital's rule by working backwards, and using the definition of derivative:

$$\begin{aligned}\frac{f'(a)}{g'(a)} &= \frac{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}}{\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}} = \lim_{x \rightarrow a} \frac{\frac{f(x) - f(a)}{x - a}}{\frac{g(x) - g(a)}{x - a}} \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} = \lim_{x \rightarrow a} \frac{f(x) - 0}{g(x) - 0} = \lim_{x \rightarrow a} \frac{f(x)}{g(x)}\end{aligned}$$

Example:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x + x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{1 + 2x} = 0$$

If it's no longer
indeterminate, then
STOP!

If we try to continue with L'Hôpital's rule:

$$= \lim_{x \rightarrow 0} \frac{\sin x}{1 + 2x} = \lim_{x \rightarrow 0} \frac{\cos x}{2} = \frac{1}{2}$$

which is wrong,
wrong, wrong!

On the other hand, you can apply L'Hôpital's rule as many times as necessary as long as the fraction is still indeterminate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - \frac{x}{2}}{x^2} \longleftarrow \frac{0}{0}$$

$$= -\frac{1}{4} \cdot \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{2}} - 1 - \frac{1}{2}x}{x^2} \quad \text{(Rewritten in exponential form.)}$$

$$= -\frac{1}{8}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2}(1+x)^{-\frac{1}{2}} - \frac{1}{2}}{2x} \longleftarrow \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{4}(1+x)^{-\frac{3}{2}}}{2} \longleftarrow \text{not } \frac{0}{0}$$

L'Hôpital's rule can be used to evaluate other indeterminate forms besides $\frac{0}{0}$.

The following are also considered indeterminate:

$$\frac{\infty}{\infty} \quad \infty \cdot 0 \quad \infty - \infty \quad 1^\infty \quad 0^0 \quad \infty^0$$

The first one, $\frac{\infty}{\infty}$, can be evaluated just like $\frac{0}{0}$.

The others must be changed to fractions first.

$$\lim_{x \rightarrow \infty} \left(x \sin \frac{1}{x} \right) \quad \longleftarrow \quad \text{This approaches } \infty \cdot 0$$

$$\lim_{x \rightarrow \infty} \frac{\sin \frac{1}{x}}{\frac{1}{x}} \quad \longleftarrow \quad \text{This approaches } \frac{0}{0}$$

We already know that $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) = 1$

but if we want to use L'Hôpital's rule:

$$\lim_{x \rightarrow \infty} \frac{\sin \frac{1}{x}}{\frac{1}{x}} = \lim_{x \rightarrow \infty} \frac{\cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right)}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \cos\left(\frac{1}{x}\right) = \cos(0) = 1$$

$$\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right) \quad \leftarrow \text{This is indeterminate form } \infty - \infty$$

If we find a common denominator and subtract, we get:

$$\lim_{x \rightarrow 1^+} \left(\frac{x-1-\ln x}{(x-1)\ln x} \right) \quad \leftarrow \text{Now it is in the form } \frac{0}{0}$$

$$\lim_{x \rightarrow 1^+} \left(\frac{1 - \frac{1}{x}}{1 \cdot \ln x + (x-1) \cdot \frac{1}{x}} \right) \quad \leftarrow \text{L'Hôpital's rule applied once.}$$

$$\lim_{x \rightarrow 1^+} \left(\frac{x-1}{x \ln x + x - 1} \right) \quad \leftarrow \text{Fractions cleared. Still } \frac{0}{0}$$

$$\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$$

$$\lim_{x \rightarrow 1^+} \left(\frac{x-1-\ln x}{(x-1)\ln x} \right)$$

$$\lim_{x \rightarrow 1^+} \left(\frac{1 - \frac{1}{x}}{1 \cdot \ln x + (x-1) \cdot \frac{1}{x}} \right)$$

$$\lim_{x \rightarrow 1^+} \left(\frac{x-1}{x \ln x + x - 1} \right)$$

$$\lim_{x \rightarrow 1^+} \left(\frac{1}{1 \cdot \ln x + x \cdot \frac{1}{x} + 1} \right)$$

L'Hôpital again.

$$\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x + 1 + 1} \right)$$

$$\frac{1}{2}$$

Indeterminate Forms: 1^∞ 0^0 ∞^0

Evaluating these forms requires a mathematical trick to change the expression into a fraction.

$$\ln u^n = n \ln u = \frac{\ln u}{\frac{1}{n}} \left\{ \begin{array}{l} \text{We can then write the} \\ \text{expression as a fraction,} \\ \text{which allows us to use} \\ \text{L'Hôpital's rule.} \end{array} \right.$$

$$\lim_{x \rightarrow a} f(x) = e^{\ln\left(\lim_{x \rightarrow a} f(x)\right)} = e^{\lim_{x \rightarrow a} \ln(f(x))} \left\{ \begin{array}{l} \text{Then move the} \\ \text{limit notation} \\ \text{outside of the log.} \end{array} \right.$$

We can take the log of the function as long as we exponentiate at the same time.

Indeterminate Forms: 1^∞ 0^0 ∞^0

Example:

$$\lim_{x \rightarrow \infty} x^{1/x} \leftarrow \infty^0$$

$$e^{\lim_{x \rightarrow \infty} \ln(x^{1/x})}$$

$$e^{\lim_{x \rightarrow \infty} \frac{1}{x} \ln(x)}$$

$$e^{\lim_{x \rightarrow \infty} \frac{\ln(x)}{x}} \leftarrow \frac{\infty}{\infty}$$

$$e^{\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1}} \leftarrow \text{L'Hôpital applied}$$

$$e^0$$

$$1$$